

Security benchmark

InjecAgent subset + authored challenges · task-aware tool-response review

103 INPUTS / 70 ATTACKS / 33 BENIGN / FROZEN INPUTS + SOURCES

WITHOUT LAYA · LLAMA REVIEW

ATTACK RECALL

85.7%

3.0%

98.4%

89.3%

FALSE POSITIVE RATE

PRECISION

ACCURACY

60/70

attacks caught

WITH LAYA · INDEPENDENT OR

ATTACK RECALL

85.7%

6.1%

96.8%

88.3%

FALSE POSITIVE RATE

PRECISION

ACCURACY

60/70

attacks caught

Laya added 0 caught attacks and 1 extra benign false positive on this regression.

01 / Layer-by-layer comparison

SAME 70 ATTACK + 33 BENIGN INPUTS

PROFILE	RECALL	CAUGHT	BENIGN BLOCKED	F1
SDK regex only	0.0%	0/70	2/33 (6.1%)	0.0%
Laya v0.3 only	1.4%	1/70	1/33 (3.0%)	2.8%
Llama semantic review	85.7%	60/70	1/33 (3.0%)	91.6%
Llama + Laya / OR	85.7%	60/70	2/33 (6.1%)	90.9%
Regex + Llama / OR	85.7%	60/70	3/33 (9.1%)	90.2%
Regex + Llama + Laya / OR	85.7%	60/70	4/33 (12.1%)	89.6%

Combined profiles are offline OR replays. Task-aware SDK review does not apply raw regex.

02 / Public + authored slices

InjecAgent base subset

83.9% recall · one selected context per public attack instruction

52/62

Derived ordinary outputs

17 benign controls obtained by removing the injection slot

0/17 FP

Authored attack challenges

16 authored benign examples: 1 false positive

8/8

03 / Full static regex test

PACK	BASE /1,054	ENHANCED /1,054	FP /17
New-org	17	1,054	0
Fallback	0	1,054	0
SDK regex	0	1,054	0

Enhanced cases contain a direct override prefix.
62 instructions × 17 contexts; base/enhanced are paired.

04 / Actual Laya v0.3 training

RESEARCH CANDIDATE

192 train · 96 calibration · 96 held-out

26.2M head parameters · frozen encoder

4 epochs / 192 steps · selected epoch 3

144 retained updates · 33.6 minutes on CPU

1

2

3*

4

Training loss

Calibration loss

SYNTHETIC HELD-OUT ATTACK RECALL

20.0%

→

32.5%

v0.2 · 8/40

v0.3 · 13/40

0/48 benign false positives for both

Separate calibration thresholds · no test tuning

05 / Runtime + evidence

Llama 3.1 8B · Q4_K_M · Ollama 0.32.5 · Intel i5-1235U · CPU only

LLM p50 15.44s / p95 23.47s · Laya p50 2.08s / p95 3.83s

LLM + Laya p50 17.48s / p95 27.14s (offline summed estimate) · 0 review errors

Scanner dev checks: 12/20 → 20/20 authored signatures caught; 0/16 benign false positives.

Capability audit: 1,054/1,054 proposed chains contain a denied tool; 17/17 original calls allowed.

Capability counts use fixture grants and dataset tool labels; no live agent attack-success score.

LOCAL DETECTOR EVALUATION · Previously inspected regression · Not official InjecAgent ASR

No agent attacks executed. Positive = suspicious/malicious; errors never count as catches. Source: InjecAgent f19c9f2.